

One More Pivot

Farsighted Search When Trials Are Scarce

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Abstract

Firms search for strategy the way Callander’s searchers probe an unknown correlated landscape: each trial is a strategy actually lived, and the last one is the one committed to. Field experiments show the search is short — roughly three quarters of startups pivot zero times in a year, a fifth exactly once, and six to eleven percent more than once — so the empirically relevant problem is not an infinite horizon truncated but a game of at most two moves, and that game has recently been solved exactly on the canonical mean-reverting landscape. This paper translates the solution into operating rules: when a struggling incumbent should pivot far, how wide to bracket before the final commitment, the explicit performance ceiling above which the grass is provably not greener, and the threshold — strictly below the performance of positions already held — at which a disappointing pivot should be abandoned rather than repaired. The abandonment rule is proved here; the other rules import from the companion exact solution. The rules are graded and falsifiable, the interior option is worth about one percent of expected value at its best, and the same field experiments that motivate the model rhyme with its central prediction: better search produces exactly one pivot, not many.

1 Callander’s problem

Callander (2011b) posed the problem this way. Agents do not know how choices map to outcomes; the mapping is the realized path of a Brownian motion, known where it has been tried, unknown elsewhere, with nearby choices performing similarly. Search is by trial and error: each agent observes the choices of earlier agents and the outcomes they realized, then makes its own choice and lives the result. Trials are not laboratory queries; each is a product shipped or a position occupied, and Callander’s own application is new product development, where the model reproduces features of the product life cycle. In the policymaking companion (Callander, 2011a), the same machinery shows that trial-and-error policymaking can become stuck at outcomes arbitrarily bad, with no institutional constraint to blame.

The workhorse of the tradition is myopic by construction, not by assumption: each agent in the sequence chooses once, for its own outcome, so nobody in the model plans two trials ahead. The survey of Bardhi and Callander (2026) lists farsighted search — a searcher who plans her few remaining trials jointly — among the tradition’s open methodological problems, and recent progress has bought tractability with restrictions: contiguous movement along a vein (Urgun and Yariv, 2025; Wong, 2025), or exact few-point placement under an information objective (Bardhi, 2024). The management literature on rugged landscapes (Baumann et al., 2019) runs on simulation for the same reason. The entrepreneur inverts Callander’s demographic structure: one agent holds the incumbent position, plans the remaining trials, and lives with the final commitment. That is the farsighted case, and the evidence of the next section says it is short.

2 The evidence: search is short

Field evidence indicates that real ventures make very few strategic pivots. In the randomized trial of [Camuffo et al. \(2020\)](#), 74% of firms pivoted zero times over roughly a year, 19% exactly once, and 7% more than once. The four-trial replication ([Camuffo et al., 2024](#)), with 759 firms, found the same shape: mode zero, 11% more than once, 6% more than twice. Practitioner folklore agrees on the mechanism — “the true measure of runway is how many pivots a startup has left” ([Ries, 2011](#)) — though no peer-reviewed cost-per-pivot measurement exists, and pivot counts should be read with the construct caution of [Kirtley and O’Mahony \(2023\)](#), who show pivots are often accumulated increments rather than single jumps.

The treatment effect is the striking part. Teaching firms a scientific approach to search raised the probability of pivoting exactly once by 8.3 percentage points, left twice unchanged, and *lowered* the probability of pivoting more than twice by 3.7 points ([Camuffo et al., 2024](#)). Better search does not produce more search. It produces one good pivot.

A searcher holding an incumbent position, one further trial, and a final commitment — at most two pivots — is therefore not a truncation of the empirically relevant problem. It is the empirically relevant problem.

3 The model, in business terms

The landscape is the exponential of a stationary Ornstein–Uhlenbeck path: log-performance mean-reverts, so there is no free lunch far from what is known, and the exponential makes rewards right-skewed, as venture outcomes are. The firm holds an incumbent position of known performance, may run one more trial anywhere, and then must commit — to the incumbent, to the tested alternative, or to an untested position between or beyond them. The payoff is the performance of the final position, not the best ever visited: the strategy committed to is the one lived with.

Mean reversion is what makes the pivot a trade. On a pure random walk landscape, moving far from the incumbent keeps its expected performance while adding variance without bound, and a skew-loving searcher would walk to the horizon; the model would recommend infinite pivots and contradict the evidence in the first sentence. Under mean reversion, distance buys variance but surrenders expected performance toward the population median, and the exact solution below prices that trade.

The exact solution is developed in a companion paper ([Cotton, 2026](#)); here we state what it says operationally.

4 Operating rules from the exact solution

Go far on bad news. With one trial and then commitment, the optimal placement has three phases. An incumbent below the median performance should be abandoned with a distant trial: its information is not worth anchoring on. A strong incumbent, above an explicit threshold, should be held without experimentation. In between, the trial goes a distance set by the mean-reversion scale of the industry, not by the firm’s ambition.

Bracket before committing. With two trials, the searcher first establishes a bracket of two known positions and then faces the signature question: commit to a tested position, or to untested

ground between them? Every interior position matches the expected performance of an explicit tested-side alternative while carrying strictly more uncertainty, so the interior is preferred exactly when the payoff is convex — when upside is what is being bought. For a risk-averse operator judged on expected utility rather than expected value, the same identity flips the verdict: the grass between two tested positions is provably browner.

The ceiling. Above an explicit performance level, the grass is not greener anywhere: mean reversion caps the expected performance of every unvisited position, and no amount of variance-loving compensates. A firm whose incumbent clears the ceiling has nothing to gain from further search, and the ceiling is a formula, not a sentiment.

A bad pivot is abandoned, not repaired. Entering a bracket is not a commitment. If the trial between two known positions comes in at or below the median, every further move inside the spoiled bracket — including repairing the disappointing position — is weakly dominated by leaving for the best outside option; strictly, if the draw is strictly below median (Proposition 1, proved in the appendix). The general rule is a threshold: the value of staying is nondecreasing in the interior draw while the exit menu does not depend on it, so the searcher stays exactly when the draw beats a cutoff (Corollary 1). In the calibration computed, the cutoff sits below the performance of the positions already held — 0.63 against anchors of 0.70 — so a pivot that lands somewhat worse than both of its neighbors is still worth developing, while one that lands badly is walked away from. The asymmetry is the model’s sharpest managerial statement.

The size of the cleverness. The interior option — the right to commit to untested ground between two tested positions — is worth about one percent of expected value at its best. Most of the value of a second trial is the trial itself, not the sophistication of where the commitment lands afterward. The model prices its own subtlety, and prices it small.

5 The rhyme with the experiments

The exact solution has a revisit window: an interval of incumbent performance in which the optimal play is one trial and a considered commitment, bounded below by the flee region and above by the ceiling. The randomized trials found that better search shifts firms toward exactly one pivot and away from both zero and many. The two shapes rhyme. We state this as a rhyme and not an identity: the treatment effect in Camuffo et al. (2024) is a nonlinearity in an intervention’s effect on pivot counts, not a causal estimate that a third pivot destroys value, and the model’s window is an interval in incumbent quality, not in pivot count. A design that measured incumbent quality at baseline could test the correspondence directly.

6 One last trial in operations

The same three-move structure appears wherever a process owner holds an incumbent operating point, has time for one more pilot, and must then deploy: the final “shot” is the deployment decision, not a laboratory run. The rules above apply verbatim, with one qualification that decides applicability: the commitment set must genuinely include untested interpolated settings within an approved operating range. If deployment is restricted to individually validated settings, the interior option is off the table and the problem reduces to the simpler tested-side rules.

For the experimental-design community the model is a benchmark rather than a rival. The trial-placement step maximizes the expected value of the best deployable setting after one more observation, which is the knowledge-gradient criterion of [Frazier et al. \(2008\)](#) with an exponential objective; no new selection principle is claimed. What the model contributes is an unusually explicit instance — exact interior maximization, an exact phase diagram — against which knowledge-gradient implementations and their approximations can be checked, and possibly undercut in computation, since on a Markov landscape a new observation changes only its containing interval.

7 A radio interpretation, tested

The mathematics does not know it is about pivots. A technician surveying WiFi signal strength holds the same problem: an incumbent mounting spot with measured power, time for one more measurement, then the equipment is mounted — possibly at an unmeasured spot between or beyond the two readings. Log-power shadowing follows the classical model of [Gudmundson \(1991\)](#): exponentially correlated in space, which is exactly the exponentiated Ornstein–Uhlenbeck landscape. On a public indoor dataset ([Feng et al., 2024](#)) — a 92-by-15 meter floor, 642 reference points, thirteen access points — the premise checks directly: near-Gaussian shadowing residuals and exponential spatial correlation with a length near seven meters, at $R^2 \approx 0.98$, in two independent reproductions (`casestudy/wifi` in the repository).

The replay is the honest part. On held-out corridors, one extra measurement lifts the mounted signal by 23% over the incumbent and the full three-shot policy by 33% — but equispaced far probing and a knowledge-gradient heuristic do better still, near 70%. The theory predicts its own defeat here: the correlation length is about eleven grid units against corridors of fifteen to ninety, so far probes are nearly fresh draws, and the small-correlation expansion of the exact formulas makes the interior option’s edge first order in the correlation while the fresh-draw variance bonus is order one. Where trials can roam many correlation lengths, measure far and take the best is leading-order optimal, and the bracket refinements of Section 4 are second-order corrections.

The business translation is the regime lesson. An industry’s correlation length decides which searcher you are. The radio surveyor roams tens of correlation lengths and should probe far and simply take the best. A venture is tethered: capabilities, customers, and identity keep the next position within roughly one correlation length of the incumbent, and that is precisely the regime — near brackets, high incumbents — where the phase diagram, the ceiling, and the abandonment threshold carry the decision. The replay’s round-one gaps are recorded with it: gains are pooled rather than conditional on the incumbent’s standing, per-corridor mean effects are unmodeled, and the flee move is truncated at corridor ends.

8 Closing

Neither the myopic workhorse nor the infinite-horizon limit describes the search the evidence shows firms conducting. The scarce-trial regime is its own theory, it now has an exactly solved case, and its rules are concrete: go far on bad news, bracket before committing, respect the ceiling, abandon bad pivots at a threshold below your anchors, and expect the cleverness premium to be small. We leave to future work the k -shot game — in the fast-reversion regime the data occupies, a singular-perturbation expansion around independent draws, with the exact three-shot solution

as its seed — along with information-motivated trials and the field test the rhyme of Section 5 invites.

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A The abandonment threshold

Work on the log scale, where a location with conditional mean μ and conditional variance ν has log-value $\mu + \nu/2$, and let $\zeta(b)$ denote the exterior two-shot value anchored at a known log-value b : explicitly $\zeta(b) = \frac{1}{2}$ for $b < 0$, $\frac{1}{2}(b^2 + 1)$ for $0 \leq b \leq 1$, and b for $b > 1$ (Cotton, 2026). A bracket has known end log-values b and d ; an interior trial has revealed log-value v .

Proposition 1 (A bad draw sends you back out). *Let $v \leq 0$, the process median or below. Then every placement interior to the two spoiled sub-brackets, and the revisit of v , is weakly dominated by exiting to $\zeta(\max(b, d))$; strictly if $v < 0$.*

Proof. Consider the sub-bracket with near end b and far end v , and an interior point with correlations λ_2 and λ to its ends, $\rho_1 = \lambda_2\lambda$. The conditional mean is $\mu = Ab + Bv$ with $A = \lambda_2(1 - \lambda^2)/(1 - \rho_1^2)$ and $B = \lambda(1 - \lambda_2^2)/(1 - \rho_1^2)$, both nonnegative, and $A \leq \lambda_2$ because $\lambda \geq \rho_1$ implies $1 - \lambda^2 \leq 1 - \rho_1^2$. Since $v \leq 0$, $\mu \leq \lambda_2 b$. The conditional variance obeys $\nu = (1 - \lambda_2^2)(1 - \lambda^2)/(1 - \rho_1^2) \leq 1 - \lambda_2^2$ by the same inequality. Hence

$$\mu + \frac{\nu}{2} \leq \lambda_2 b + \frac{1 - \lambda_2^2}{2} \leq \sup_{\ell} \left[\ell b + \frac{1 - \ell^2}{2} \right] = \zeta(b),$$

the last identification holding for every real b : for $b \in [0, 1]$ the supremum is $(b^2 + 1)/2$ at $\ell = b$; for $b > 1$ it is b at $\ell = 1$; for $b < 0$ it is $\frac{1}{2}$ at $\ell = 0$. For $v < 0$ any strictly interior point has $B > 0$, so $\mu < \lambda_2 b$. The revisit of v pays $v \leq 0 < \frac{1}{2} \leq \zeta$. The same bound applies to the other sub-bracket with d in place of b . $\square$

Proposition 2 (Monotone value). *For the k -shot game with any frontier of observed location–value pairs, the optimal value is nondecreasing in each observed value.*

Proof. By induction on k . Every candidate placement’s conditional mean — bridge or exterior ray — is a nonnegative linear combination of observed values, and every conditional variance depends on locations only. At $k = 0$ the payoff is a maximum of terms nondecreasing in each observed value. For the step, raise one observed value and couple the reveal at any placement upward by the shift in its conditional mean; the inductive hypothesis and the maximum over placements preserve monotonicity. $\square$

Corollary 1 (Threshold rule). *After an interior reveal v with one shot left, the exit menu is anchored at outer values and does not depend on v , while the stay menu is nondecreasing in v by Proposition 2. A nondecreasing function crosses a constant once: the searcher stays exactly when v exceeds a threshold $v'(b, d, \rho)$. At $(b, d, \rho) = (0.7, 0.7, 0.5)$ the threshold computes to 0.6301, below both anchors; the computation is `verify/check_commitment.py` in the repository.*